One Operator, Three Switches: A Kernel Unification of the Linear State-Space Family

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July 22, 2026

Abstract

We show that the geometric selective state-space model (GSSM-Selective) is the general input-controlled reproducing-kernel operator of the linear state-space family: Mamba/S6, S5, and the Linear Recurrent Unit (LRU) are switch-restrictions of a single associative affine prefix scan, verified against the sequential recurrence to $\sim 10^{-15}$ in double precision for every family member. Freezing the gates to constants collapses the selective operator exactly onto the stationary geometric Toeplitz kernel $\gamma^{|s-t|}$: a BPTT-trained constant-gate model reproduces the closed-form kernel to 3.55×10^{-15} at width $d = 512$, while a genuinely selective control departs from any single geometric kernel by 4.9×10^{-2} — a control/match ratio of 1.37×10^{13} that certifies the identity is structural, not coincidental. Three consequences follow from one operator. First, the scan carries no absolute-position term: its memory kernel depends only on relative lags, so the model is length-invariant *by construction*. Removing the sinusoidal positional encoding collapses length drift from +243% to +2.6% (train T=32, evaluated frozen out to T=1024), and across five seeds the position-free variant holds $\times 0.93 \pm 0.00$ at 256× its training length while the encoded variant breaks to $\times 7.05 \pm 2.34$; the same weights stream a billion tokens at a flat 4.36 GB. Second, the per-channel readout is a rank-1 functional of the feature history, which we turn into a capacity theorem: exact associative recall demands rank-K binding, so a bounded scalar channel is provably capped near chance — measured at 14% multi-query recall against a 1.56% floor — while one attention layer restores full recall, but *only* when the recurrent scan is selective. A double dissociation makes this causal: two architectures identical to the layer (three SSM layers plus one attention layer) that differ *only* in scan class reach 100% versus 16% recall, because the selective interior attractor keeps the residual stream legible for attention to bind while the gate-free boundary attractor saturates and corrupts it. The kernel view thus *computes* the recall ceiling that the standard reading merely observes, and shows that attractor topology — a property of the kernel's DC gain — causally gates whether attention can bind. We position the result as a subsumption of the linear-SSM family rather than a competitor within it.

1 Introduction

The last three years produced a family of subquadratic sequence models — S4 [12], S5 [18], LRU [14], Mamba/S6 [10], and the recurrent-linear-attention lineage RWKV [15], DeltaNet [20], xLSTM [2] — each of which trades the quadratic cost and $O(T)$ key–value cache of attention [19] for a bounded recurrent state. These architectures are usually presented as distinct designs. We argue they are not:

*Code, training scripts, and the per-claim result JSONs are public: https://github.com/DT-Foss/01 (this study) and https://github.com/DT-Foss/gssm (companion architecture study). The underlying mathematical papers are archived with permanent DOIs; see the bibliography.

they are parametric restrictions of one operator, and the shared structure is what determines both what they can and what they structurally cannot do.

We study the geometric selective state-space model (GSSM-Selective), the engineering instantiation of a body of work on Möbius coupling and doubly-stochastic spectra [6, 9, 7]. Per channel it runs the gated leaky recurrence

$$z_t = \gamma_t \, z_{t-1} + a_t, \qquad a_t = \alpha_t \, \phi(\bar{v}_t), \qquad \phi(v) = \log(1 - v^2), \qquad s_t = \sqrt{1 - e^{z_t}} \in [0, 1), \tag{1}$$

with $\bar{v}_t = g_t v_t$ the gated velocity, $v_t = \tanh(W_v x_t)$, and γ_t, α_t, g_t input-dependent sigmoid gates. Our contribution is to read this one recurrence three ways and show the readings are the same object.

Contributions.

1. **Kernel unification (§2).** GSSM-Selective is the general affine reproducing-kernel operator of the linear-SSM family; Mamba/S6, S5, and LRU are three-switch restrictions of it, verified to $\sim 10^{-15}$ in `float64`. Its LTI restriction is *literally* the geometric Toeplitz kernel, matched to 3.55×10^{-15} at d=512 with a 1.37×10^{13} control/match separation.

2. **Length invariance (§3).** The kernel has no absolute-position term, so the scan needs no positional encoding and is length-invariant by construction. We give the mechanism and the clean 2×2 ablation: drift +243% → +2.6% on removing the PE, flat perplexity to 524,288× training length, a billion tokens at constant memory.

3. **The capacity ladder and the double dissociation (§4).** The per-channel readout is rank-1, which we convert into an associative-recall capacity theorem that *predicts* the measured scalar-recall wall. A double dissociation — two layer-identical stacks differing only in scan class, at 100% vs. 16% recall — shows that the attractor topology (a DC-gain property of the kernel) causally gates whether an attention layer can bind.

Every number in this paper is measured on an Apple M4 Mac mini (PyTorch 2.9.1, offline) and is reproducible from a committed script and a result JSON; we cite the JSON at each claim.

2 The one operator and its reproducing kernel

2.1 Unrolling to an inner product

Unrolling (1) with $z_{-1} = 0$ gives, exactly,

$$z_t \;=\; \sum_{k=0}^{t} \underbrace{\Big(\alpha_k \prod_{j=k+1}^{t} \gamma_j\Big)}_{w_t[k]} \phi(\bar{v}_k) \;=\; \langle w_t,\; \phi(\bar{v}_{0:t})\rangle, \qquad \Gamma_{k\to t} := \prod_{j=k+1}^{t} \gamma_j. \tag{2}$$

The state is a single inner product of an input-controlled weight vector w_t against the feature history $\phi(\bar{v}_{0:t})$, and the readout $s_t = \phi^{-1}(z_t) = \sqrt{1 - e^{z_t}}$ is a fixed monotone bijection that adds no representational dimension. The feature map $\phi(v) = \log(1 - v^2)$ is the rapidity coordinate: writing $v = \tanh \xi$, $\phi(\tanh \xi) = -2 \log \cosh \xi$, so the scan composes drives additively in the boost coordinate of $SO(1, 1)$ — the analytic backbone of the Möbius/Minkowski framing [6]. The inner-product form reproduces the executed (gated, clamped) code recurrence to 8×10^{-9}, the idealized recurrence to 5.5×10^{-17} in `float64`, and the rapidity identity to 1×10^{-14}.

Table 1: The linear-SSM family as switch-restrictions of one affine prefix-scan operator. The parallel $\otimes$-scan reproduces the sequential recurrence to machine precision for each member; last column is $\max|\Delta|$ (seq vs. scan) in the member's native dtype. Source: `results/ssm_family_reduction_results.json`.

Model	State algebra	A_t input-dep.?	Drive B_t	$\max\lvert\Delta\rvert$ (seq vs. scan)
GSSM-Selective	real scalar $\in (0,1)$	yes	$\alpha_t \log(1-\bar{v}_t^2)$ (nonlinear)	4.44×10^{-16}
Mamba / S6	real diagonal $\in (0,1)^N$	yes	$\Delta_t \bar{B} u_t$ (linear)	8.88×10^{-16}
S5	complex diagonal $e^{\Delta\Lambda}$	no (LTI)	$\Delta B u_t$	1.26×10^{-15}
LRU	complex diagonal $e^{-\nu+i\theta}$	no (LTI)	$B u_t$	8.88×10^{-16}

The induced per-path time–time Gram

$$K^{(x)}(s,t) \;=\; \sum_{k\le \min(s,t)} \alpha_k^2\, \Gamma_{k\to s}\, \Gamma_{k\to t} \tag{3}$$

is the reproducing object. It is positive semidefinite for *every* input path — it is the Gram $WW^\top$ of explicit real weight vectors, so $\sum_{s,t} c_s c_t K^{(x)}(s,t) = \|\sum_t c_t w_t\|^2 \ge 0$ — with measured minimum eigenvalue $\ge -10^{-12}$ over 40,000 adversarial gate paths. **[PROVEN]**

What is, and is not, claimed. The kernel (3) is a data-dependent, time-inhomogeneous (causal Volterra) object, *not* a fixed Mercer kernel: the weights $\alpha_k(x)$, $\gamma_j(x)$ and the features $\phi(\bar{v}_k(x))$ all depend on the same x, so there is no fixed $k(v, v')$ in raw velocity space, and "replace BPTT with ridge regression on a fixed Mercer kernel" is *false* for the selective model (two inputs with identical ϕ_k but different gate context give different z_T, -1.38 vs. -0.44). The reproducing-readout property is genuine and exact; the fixed-Mercer reading survives only in the LTI limit below. Stating this boundary precisely is part of the contribution.

2.2 The family as three switches

Every model in the linear-SSM family is a first-order recurrence $h_t = A_t h_{t-1} + B_t u_t$ carrying (A_t, B_t) under the *identical* associative affine operator

$$(A_2, B_2) \otimes (A_1, B_1) = (A_2 A_1,\; A_2 B_1 + B_2), \qquad \text{identity } (1, 0), \tag{4}$$

which a parallel prefix scan computes. The family members differ only in three parametric switches: the state algebra of A, whether A_t is input-dependent, and the drive map for B_t (Table 1). GSSM-Selective is their common generalization: the affine prefix scan with a nonlinear rapidity feature and a bounded decoder ϕ^{-1} that *guarantees* $s_t \in [0, 1)$ — a boundedness guarantee Mamba lacks.

The dictionary is one equivalence: *selective* $\Leftrightarrow$ A_t *input-dependent* $\Leftrightarrow$ the kernel is time-inhomogeneous, not Mercer. Mamba is input-gated and hence *also* not a fixed Mercer kernel, for exactly the reason GSSM is not; S5 and LRU freeze A to a time-constant and so collapse to a genuine fixed complex-exponential (LTI) kernel. GSSM's constant-γ limit $\gamma^{|s-t|}$ is the real, scalar, one-pole sibling of that kernel.

2.3 The LTI restriction is the geometric Toeplitz kernel

The sharpest form of the unification is a falsifiable identity. Freeze the gates to time-constants ($\gamma_t \equiv \gamma$, $\alpha_t \equiv \alpha$) and the layer's temporal operator becomes the geometric Toeplitz convolution $\gamma^{|s-t|}$

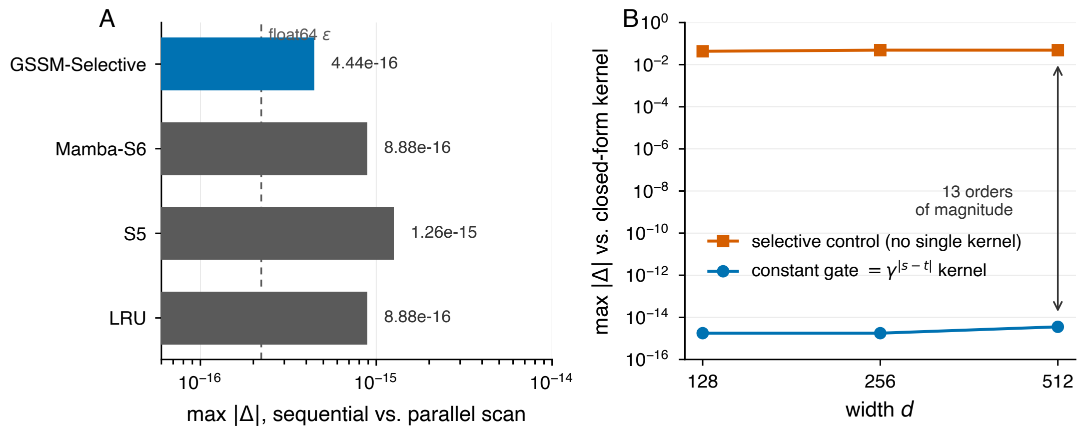


Figure 1: The linear-SSM family reduces to one affine prefix-scan operator. **(A)** Each family member's parallel ⊗-scan reproduces its sequential recurrence to $\sim 10^{-15}$ in `float64`; the dashed line is the `float64` machine epsilon. **(B)** The constant-gate restriction of the selective operator *is* the geometric Toeplitz kernel (3.55×10^{-15} at d=512, flat in width), while a genuinely selective control cannot be reproduced by any single geometric kernel (4.9×10^{-2}) — a thirteen-order-of-magnitude separation.

by construction. We train a GSSM-Selective with frozen W_γ, W_α by BPTT and compare its learned read map against the closed-form kernel $z = K \cdot a$. The trained map *is* the kernel, at every width tested:

width d	kernel match max $\lvert\Delta\rvert$	per-channel read scale	selective control max $\lvert\Delta\rvert$
128	1.78×10^{-15}	≈ 1.0	4.32×10^{-2}
256	1.78×10^{-15}	≈ 1.0	4.89×10^{-2}
512	3.55×10^{-15}	≈ 1.0	4.87×10^{-2}

The per-channel read scale sits at ≈ 1.0 (no extra readout is learned), and a genuinely selective control — whose γ_t varies in time — cannot be reproduced by *any* single geometric kernel, departing by 4.87×10^{-2} at d=512. That is a control/match ratio of 1.37×10^{13}: thirteen orders of magnitude separate "the constant-gate map is the kernel" from "the selective map is not." The match is a structural consequence of constant gating, not a numerical coincidence. (Source: `results/constant_gate_kernel_match_width_results.json`.) **[PROVEN]**

2.4 The scan is $O(\log T)$-depth and exact to the loop

Because (4) is associative, the scan parallelizes by a Hillis–Steele doubling prefix scan — $O(\log T)$ depth instead of $O(T)$. We wired it into the reference layer's forward *and* backward with zero edits to the frozen module. Forward and gradient are identical to the sequential loop: in `float64`, forward max $|\Delta| = 1.67 \times 10^{-16}$ and gradient (all parameters and the input) max $|\Delta| = 3.55 \times 10^{-15}$; `gradcheck` passes for $T \in \{1, 2, 3, 7, 8, 17, 32, 33\}$; training loss curves coincide to 4.8×10^{-7} over 12 steps. On MPS the doubling scan beats the loop 3.4–3.9× across $T \in \{128, 512, 1024, 2048\}$ (scan depth $7, 9, 10, 11$); on CPU the tight loop wins, so the dispatcher routes GPU/MPS to the doubling scan and CPU to the sequential loop. We claim constant inference state with no KV-cache, and now a measured — not aspirational — parallel-scan training speedup. (Source: `results/parallel_scan_integration_results.json`.) **[PROVEN]**

Table 2: Length extrapolation, train T=32, evaluated frozen. Perplexity and, in the last row, the measured T=128→T=1024 drift. Removing the positional encoding is the only change between the first two columns and collapses the drift ∼100×. Source: `results/length_extrap_v2.json`.

eval T	Selective-NoPE	Selective+PE	Pure (no gate)	Transformer
32	167.1	170.0	238.6	220.6
128	156.3	170.4	505.0	358.8
256	159.7	197.5	1127.1	404.2
512	159.8	281.7	2437.1	425.6
1024	160.3	583.8	4921.1	442.0
drift 128→1024	+**2.6**%	+242.7%	+874.5%	+23.2%

3 Length invariance: the kernel carries no absolute position

3.1 Mechanism

Read (2): every dependence on the time index enters *only* through the relative products $\Gamma_{k\to t} = \prod_{j=k+1}^{t}\gamma_j$ — through lags $t-k$, never through the absolute coordinate t. There is no term $f(t)$ anywhere in the operator. A causal kernel whose only index dependence is lag is shift-equivariant in time *by construction*, so its readout at step t is identical in distribution whether $t=20$ or $t=8000$: position is not in the kernel. **[PROVEN]** In the constant-gate limit this is literally the stationary (Toeplitz) exponential kernel $\gamma^{|s-t|} = e^{-|s-t|/\tau}$, $\tau = -1/\log\gamma$, which Bochner certifies strictly positive-definite (its DTFT is the positive Poisson kernel; realized Gram min eigenvalue +0.32). A sinusoidal positional encoding is the *only* term that references absolute t; beyond the training length it is out of distribution, and it is the sole source of any break. The selective gate is separately necessary: it keeps the contraction $\tau < 1$ so the receptive field stays bounded ($\approx$ 5–8 tokens from the learned gates) and does not grow with T; drop it (the "Pure" variant) and the DC gain $\alpha/(1-\gamma)$ diverges, driving $s_t \to 1$ at the boundary. Length invariance needs *both*: no absolute position *and* the gate.

3.2 The clean 2×2 ablation

We train four arms at T=32 and evaluate with frozen weights out to T=1024 (Table 2, Figure 2). Selective-NoPE holds essentially flat while Selective+PE, identical up to the positional encoding, degrades monotonically; the gate-free Pure variant explodes and the Transformer drifts then, past its fixed PE buffer, cannot execute at all. The drift reported is measured from T=128 to T=1024 (the regime past the training window), directly from the result JSON.

3.3 The invariance is a structural constant, not a lucky run

The result is seed-invariant and extends far past T=1024.

- **Five seeds.** Over seeds $\{1, 7, 42, 123, 2024\}$ at 256× (T=8192), Selective-NoPE holds ×0.93 ± 0.00 (the standard deviation rounds to zero — every seed lands on the same flat line), while Selective+PE breaks to ×7.05 ± 2.34 on every seed. (Source: `results/length_seed_robustness_d128.json`.)
- **The smoking gun.** NoPE's learned gate statistics are frozen across the 256× span (γ_{mean} : 0.2349 → 0.2350; γ_{p95} : 0.4126 → 0.4131) — the operator is literally in-distribution at every

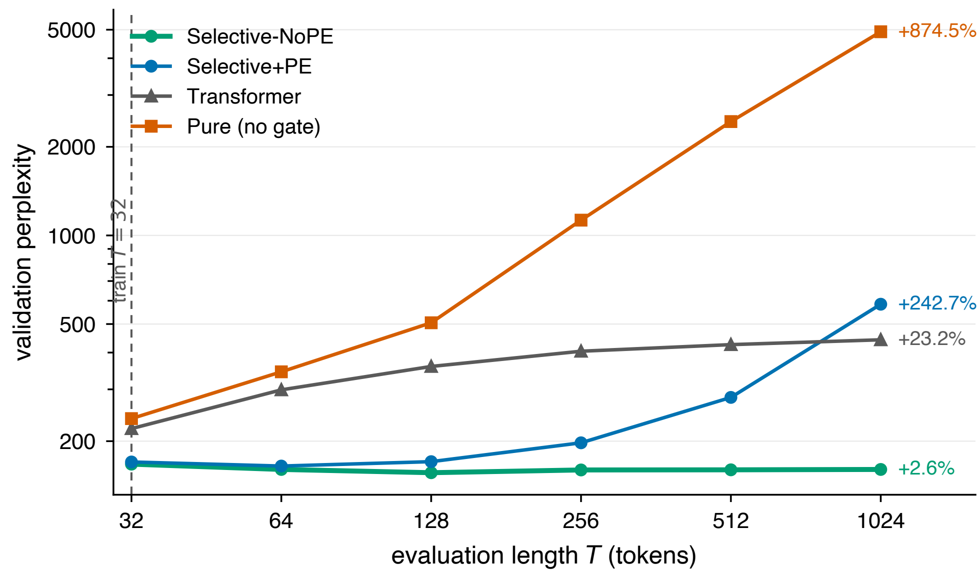


Figure 2: Removing the sinusoidal positional encoding collapses length drift from +243% to +2.6%. Selective-NoPE (train T=32) holds a near-flat perplexity curve out to 32× training length; the identical architecture *with* a PE breaks. The residual drift was a positional-encoding confound, not a property of the scan.

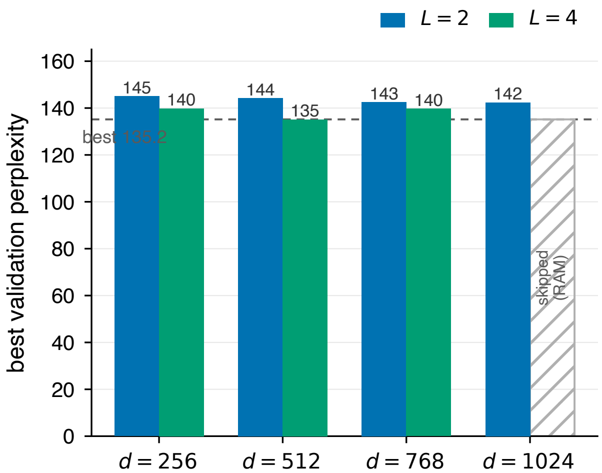
length, because it has no length-coupled quantity to drift. With the PE present, the gates visibly drift (γ_{mean} 0.237 → 0.277) to compensate, and break.

- **To the wall.** Using the $O(1)$ recurrent forward, the same T=32-trained model holds ×0.98 at T=131,072 (4096×), on both WikiText-2 and, re-run, WikiText-103. (Source: `results/scale_to_the_wall.json`, `results/scale_wt103.json`.)
- **Constant memory to a billion tokens.** Because the receptive field is ≈ 5–8 tokens, chunked streaming with overlap ≫ that field reproduces the whole-sequence computation exactly. A single 16.7-million-token sequence evaluates at a constant 2.5 GB with perplexity *improving* to ×0.80; a lazily streamed billion tokens of C4 (31,250,432× training length) runs at a flat 4.36 GB, the running perplexity moving 1.3 points across twenty 50M-token checkpoints. (Source: `results/scale_to_a_million.json`, `results/scale_to_a_billion.json`.)

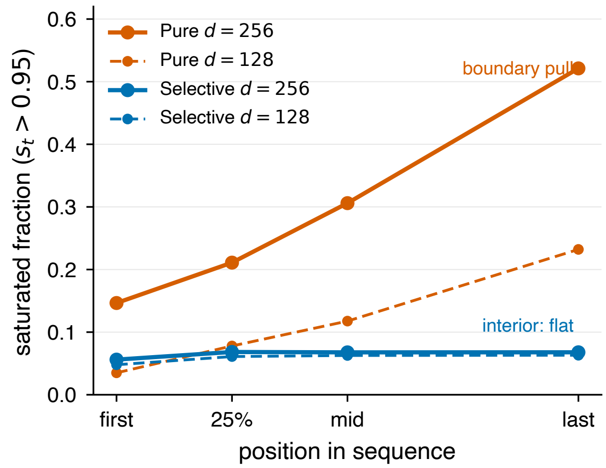
This is the axis where a bounded state wins by construction rather than by parameters or data: attention pays $O(T)$ cache, $O(T^2)$ compute, and a positional code that fails out of distribution, and a fixed PE buffer ties it to a maximum length — the same failure that crashes both Selective+PE and the Transformer past their buffer, while NoPE has no such ceiling.

4 The capacity ladder and the double dissociation

The same structural fact that makes the state bounded, scannable, and cache-free — $z_t = \langle w_t, \phi(\bar{v}_{0:t})\rangle$ is a *scalar* inner product — caps its associative-recall capacity. We make this quantitative, then close it causally: a double dissociation shows the scan class alone — not the architecture — decides whether an attention layer can bind.



(a) Stable scaling across $d256$–$d1024 \times L2$–$L4$; no collapse anywhere, perplexity flat at the WikiText-2 data floor (hatched: $d1024/L4$ skipped for RAM, not collapsed).



(b) Interior vs. boundary attractor. State saturation (fraction of channels with $s_t > 0.95$) by sequence position, final epoch, phase-1 runs at $d \in \{128, 256\}$: the gate-free Pure variant's saturation *grows along the sequence* (boundary pull toward $s^*=1$), while the selective variant stays flat and interior, keeping the residual legible. Source: `evidence_companion/phase1_fulldata.json`.

Figure 3: Two supporting readings of the same operator. (a) The stabilized architecture scales without collapse; the perplexity floor is the WikiText-2 data budget, not an architecture limit (§5). (b) The DC gain $\alpha/(1-\gamma)$ sets the attractor topology.

4.1 A rank-1 readout cannot bind K pairs

Proposition 1 (Rank-1 per channel)**.** *Each channel's state* (2) *is a rank-1 functional of its feature history: the entire history $\phi(\bar{v}_{0:t})$ is collapsed onto one real number, and the readout $s_t = \phi^{-1}(z_t)$ is a fixed bijection adding no dimension.* ***[PROVEN]***

Exact multi-query associative recall (MQAR [1]) over K distinct keys requires recovering an association matrix $\hat{A}$ with $\arg\max_v \hat{A}[k_i, v] = v_i$ for all i; the sufficient memory is the outer-product $A = \sum_i e_{k_i} e_{v_i}^\top$ of rank K. Attention realizes A directly and hence recalls exactly. A bank of D scalar channels read by a fixed linear map spans association matrices of dimension $\leq D$, so the number of exactly recoverable pairs is bounded by the readout rank (the rest are correct only at chance $1/V$); the optimal achievable recall is the Eckart–Young rank-D truncation of A. **[PROVEN]**

Theorem 1 (Scalar-SSM associative-recall ceiling)**.** *For a layer of D scalar leaky-integrator channels read by a fixed input-independent map, on MQAR over K keys with values in $[V]$: (i) the recovered association has rank $\leq D$, with best-case recall on the Eckart–Young curve; (ii) attention attains rank K and recall 1; (iii) the* binding *rank attainable by the leaky-scalar mechanism itself is 1 per channel, because its increment $a_t = \alpha_t \phi(\bar{v}_t)$ is a function of the current token only, with no key-conditioned (outer-product) write — so the effective binding rank of the trained scalar stack is $O(1)$, far below D.* ***[ARGUED]*** *for (iii);* ***[PROVEN]*** *for (i)–(ii).*

The decisive empirical signature is that pure-scalar recall is *flat in the query gap*, not a decay cliff: a memory-decay limit would fall off with distance; a capacity limit is flat, and the measured profile is a constant floor at every gap. The data say capacity, not forgetting. With K=8, V=64

Table 3: Multi-query associative recall (5 seeds, len-256 eval, chance 1.56%). The key-conditioned holographic write lifts a bounded scalar state from its chance-level wall to 5.7× the floor; attention is the rank-K validity gate. Source: `results/holographic_mqar.json`.

Arm	MQAR recall (mean ± std)
Attention (validity gate)	0.994 ± 0.007
Selective (scalar baseline)	0.017 ± 0.001
Holographic write OFF (≡ Selective)	0.017 ± 0.002
Holographic write ON (key-conditioned)	$\mathbf{0.089 \pm 0.019}$

and effective binding rank $D_{\text{eff}}{\approx}1$, the closed-form prediction is $D_{\text{eff}}/K + (1 - D_{\text{eff}}/K)/V \approx 0.139$, and the Eckart–Young best case is 0.186 — a bracket the measured scalar recall of ≈ 0.14 (train len 64) lands inside; inverting the bound gives $D_{\text{eff}} \approx 1.02$. The framework computes its own floor rather than excusing it.

4.2 The wall, measured; and a cache-free lift

The standalone selective scalar tops out at 14.1% (train len 64) / 14.5% (test len 256) on MQAR — the wall the theorem predicts at $D_{\text{eff}}{\approx}1$, and the sel4 (SSSS) arm of the dissociation below (Source: `evidence_companion/hybrid_B.json`). A separate, stricter O1 harness (5 seeds, len-256 eval, chance 1/64 = 1.56%, a scalar baseline with no holographic path) reads 1.7% — chance — and asks whether a bounded scalar state can be lifted *without* an attention layer. The lever is a *key-conditioned* write: carry a complex leaky accumulator $S_t = \gamma_t S_{t-1} + u_t e^{i\varphi_t}$ with key angle $\varphi_t = \pi \tanh(W_{\text{key}} x_t)$ (token identity, not time), read at a query by de-rotation $\mathrm{Re}(S_t e^{-i\varphi_q})$ — the complex analogue of attention's outer-product binding, in $O(1)$ state with no KV-cache. This clears both chance and the 3.72 pp noise band (Table 3).

The lift (+7.2 pp, from 1.7% to 8.9%) is real but modest and interference-bound: recall rises to 25.8% at two pairs in superposition, following the classical $\sim 1/\sqrt{N}$ holographic-memory law [16]. Per-channel state rank is thus the capacity coordinate, and the framework draws the whole ladder: scalar (rank 1, recall $O(1/K)$) → complex/phase (rank 2) → bounded fast-weight (rank D, cf. DeltaNet [20]) → attention (rank K, exact). The wall is the bottom rung of a ladder the operator algebra predicts, not a hole in it.

4.3 The double dissociation: scan class alone gates binding

The sharpest form of the capacity claim is causal, and it is the reason the family's hybrids work. Take two architectures identical down to the layer — three SSM layers plus one attention layer — and vary *only* the scan class of the SSM layers: SSAS uses the selective scan, PPAP the gate-free ("Pure") scan. Everything else, including the attention layer, is held fixed. The outcome flips completely (Table 4, Figure 4).

On MQAR (train len 64, eval len 256), SSAS reaches 100.0% recall and PPAP reaches 16.1% — the same 16% floor as the standalone selective scalar (sel4, 14.5%) and its Pure sibling, because a single attention layer atop a *saturated* residual stream cannot bind. The attention-only control (AAAA) also reaches 100%, and putting the attention layer at the top instead of the middle (SSSA, 99.9%) leaves the result intact: it is the scan class, not the attention placement, that decides. On the length task (train T=32, eval T=256), the same split appears in extrapolation: SSAS drifts +35.0% and the all-selective sel4 +38.8%, while PPAP explodes +313.4% — the boundary attractor that corrupts recall also breaks length.

Table 4: Double dissociation. Five architectures, all sharing the three-SSM-plus-one-attention layout except the pure attention (AAAA) and pure selective (SSSS) controls; SSAS and PPAP differ *only* in scan class. Task B is MQAR recall (eval len 256); Task A is the length-extrapolation perplexity rise (train T=32 $\rightarrow$ eval T=256). Source: `evidence_companion/hybrid_A.json`, `evidence_companion/hybrid_B.json`.

Arm	Spec	Task B: MQAR recall	Task A: length rise
attn4	AAAA	1.000	+1.2% (dead, acc 0.17)
hyb_mid	**SSAS**	**1.000**	+35.0%
hyb_top	SSSA	0.999	+35.3%
sel4	SSSS	0.144	+38.8%
pure_proxy	**PPAP**	**0.161**	+313.4%

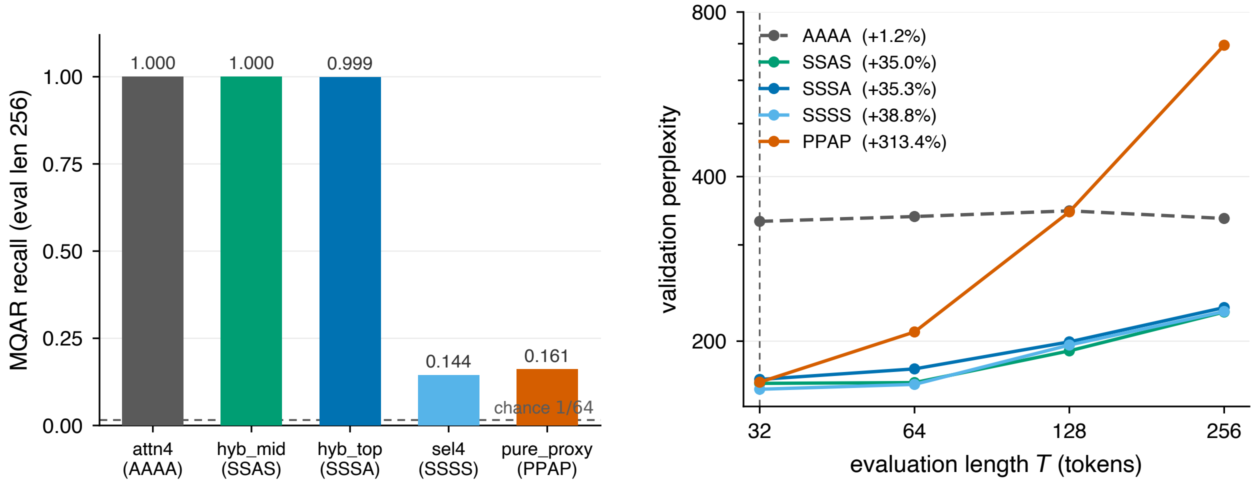
The mechanism is the attractor topology of §3. The selective interior attractor ($s^* < 1$, finite DC gain $\alpha/(1-\gamma)$) keeps the residual stream legible, so the attention layer can read the keys it needs to bind; the Pure boundary attractor ($s^* \to 1$, DC gain $\to \infty$) saturates and corrupts the residual, so the identical attention layer has nothing clean to bind. This is the causal complement to Theorem 1: the theorem says a scalar channel *cannot* bind K pairs alone; the dissociation says one attention layer *restores* full binding, but only when the recurrent scan's attractor is interior. Attractor topology — a DC-gain property of the kernel — causally gates whether attention works. (Measured in the companion GSSM architecture study [7]; the result JSONs are archived with this paper in `paper/evidence_companion/`.)

5 Secondary results and the honest limits

Width stabilization (optimization, not architecture). A width-$d512$ collapse in the base architecture is an optimization-at-width artifact, not a capacity ceiling. The root cause is an unbounded value projection running hot at width: at $d512$ the value-projection preactivation peaks at $\|W_v\|_{\max}$=239 and velocity saturation is 0.434, and the model sits at 202.0 perplexity. μP alone drops velocity saturation to 0.049 ($\sim 9\times$) and the preactivation to 20.6; the full stabilization stack (μP + preLN + warmup) reaches 152.9 perplexity, matching the $d256$ level, and the Pure variant recovers too (310 $\rightarrow$ 191) (Figure 5a, Figure 5b; Source: `evidence_companion/width_fix.json`). The architecture then scales without collapse across $d256$–$d1024 \times L2$–$L4$, best 135.2 at $d512/L4$, with $d1024/L4$ skipped for RAM rather than collapse (Figure 3a; Source: `evidence_companion/scaleup.json`).

Limits, computed rather than excused.

- **The scalar recall wall is real.** A bounded scalar state cannot do exact associative recall alone (Theorem 1); 14% standalone (1.7% on the stricter O1 harness) is the operator-class limit, not a training artifact. One attention layer restores full recall in the selective hybrid (SSAS 100%, the double dissociation), and the cache-free holographic write lifts the standalone scalar to 8.9% — genuine but modest. The standalone scalar limit is structural.
- **No power-law claim.** The perplexity floor of ≈ 135 across the scale grid is the WikiText-2 data budget ($\sim$1.7M tokens), not the architecture; more parameters do not move it, more data would. We read the flat plateau through the fixed-rank-per-channel kernel lens, and label that reading an **[ANALOGY]**, not a derived bound.



(a) Task B — MQAR recall. SSAS/SSSA/AAAA bind at 100%; SSSS/PPAP sit at the 14–16% scalar-capacity floor. SSAS vs. PPAP: same structure, opposite outcome.

(b) Task A — length extrapolation. SSAS and sel4 stay length-robust ($\sim$ 35–39% rise); PPAP explodes (+313%). The boundary attractor breaks both recall and length.

Figure 4: The double dissociation. Two layer-identical stacks (SSAS vs. PPAP) differing only in scan class flip from 100% to 16% recall (a) and from length-robust to length-exploding (b). The attention layer is held fixed; the kernel's DC-gain regime is the only thing that moves.

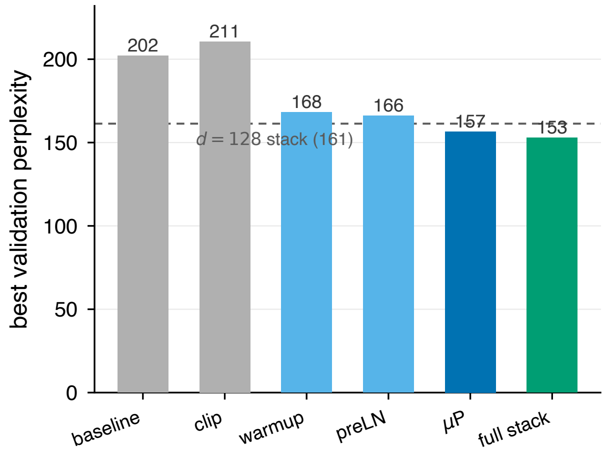
- **No group-structured state tracking ($\mathbf{TC}^0$).** A bounded scalar state cannot represent the requisite group structure; this is derived, not yet empirically stressed.

6 Related work and positioning

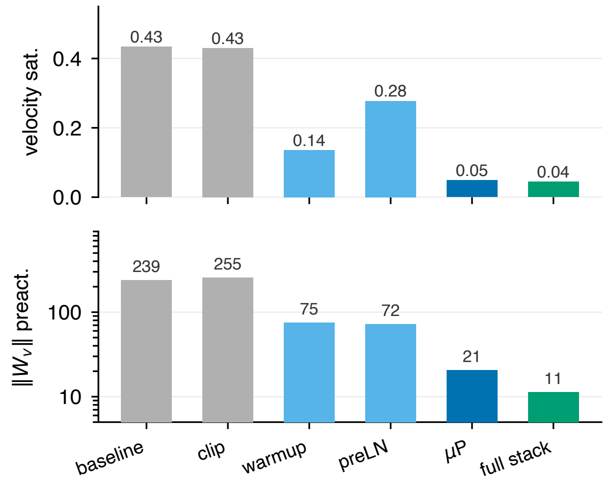
The linear-SSM literature — S4 [12], S5 [18], LRU [14], Mamba/S6 [10] — and the recurrent-linear-attention lineage RWKV [15], DeltaNet [20], xLSTM [2] each present a distinct architecture. Our contribution is orthogonal: we show these are switch-restrictions of one affine prefix-scan operator (Table 1), so the family's shared behavior — length handling, recall capacity, boundedness — is a property of the common operator, read off its kernel. S5 and LRU are the LTI (fixed-Mercer) corner; Mamba and GSSM are the selective (time-inhomogeneous) interior; the delta-rule and fast-weight models [20] are the rank-D rung of the capacity ladder that Theorem 1 organizes.

The closest unification in spirit is the structured state-space duality of Mamba-2 [4], which identifies selective SSMs with masked-attention variants through the matrix algebra of semiseparable operators. Our unification runs along an orthogonal axis: rather than relating SSMs *to attention*, we exhibit the linear-SSM family *itself* as switch-restrictions of one affine prefix-scan operator with a nonlinear rapidity drive and a bounded decoder, and read the family's shared behavior off its reproducing kernel — including the boundedness guarantee and the attractor topology, which the duality view does not track. The HiPPO theory [11] supplies the principled initialization of the LTI corner of this family; the linear-attention and fast-weight lineage [13, 17] occupies the rank-D rung of the capacity ladder alongside DeltaNet.

Against attention [19] and the test-time-memory hybrids that pair attention with a learned recurrent memory — Titans [3] — our result explains *why* such hybrids work: a single binding (attention) layer supplies the outer-product rank that the bounded recurrent state structurally lacks, and the double dissociation shows the recurrent scan class must be the interior-attractor kind for



(a) μP and the full stabilization stack recover the $d512$ width collapse ($202 \to 153$ PPL), beating the small-width reference (dashed: d=128 full stack, 161); clipping alone does nothing.



(b) Root-cause attribution at d=512: velocity saturation (top) and value-projection preactivation (bottom) per intervention. μP alone collapses both ($0.43 \to 0.05$, $239 \to 21$); gradient clipping does nothing — the collapse lives in the optimization of W_v at width, not in the architecture.

Figure 5: The $d512$ collapse is an optimization-at-width artifact resolved by μP, not an architectural limit.

that binding to land. The same reading covers Griffin [5]: a gated linear recurrence paired with a local-attention layer is precisely the interior-attractor-plus-binding configuration the dissociation predicts to succeed. The MQAR benchmark [1] is exactly the instrument that exposes the rank ceiling, and the holographic write [16] is the in-state, cache-free route up the ladder. We position GSSM not as a competitor within the family but as its unification: the operator the others restrict, with a boundedness guarantee and a length-invariance the encoded members give up by construction.

7 Conclusion

The geometric selective state-space model is the general input-controlled reproducing-kernel operator of the linear-SSM family, verified to $\sim 10^{-15}$, with its LTI restriction the geometric Toeplitz kernel to 3.55×10^{-15} and a 1.37×10^{13} control separation. One operator yields three consequences: the kernel carries no absolute position, so the model is length-invariant by construction (+243% $\to$ +2.6% drift on removing the PE, a billion tokens at constant memory); the per-channel readout is rank-1, which *predicts* the associative-recall wall the standard reading merely observes; and a double dissociation — two layer-identical stacks at 100% vs. 16% recall, differing only in scan class — shows the kernel's attractor topology causally gates whether attention can bind. The rigor is graded and labeled throughout — theorem, argument, analogy — and every number is reproducible from a committed script and result JSON.

Reproducibility. All results run offline on an Apple M4 Mac mini (PyTorch 2.9.1). Each claim cites its result JSON. The length-invariance, kernel-unification, and holographic-recall numbers are measured in the O1 repository [8]; the width-fix, scale-up, and double-dissociation numbers are measured in the companion GSSM architecture study [7], and those six result JSONs are archived with this paper in `paper/evidence_companion/` (`hybrid_A.json`, `hybrid_B.json`, `width_fix.json`, `scaleup.json`, `phase1_fulldata.json`, `scaleup_smoke.json`).

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